



23<sup>rd</sup> - 24<sup>th</sup> July 2025

# International Conference on Engineering and Advanced Technology 2025

## Sponsor



## Organizers



## Partners



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## Sponsors, Organizers, and Partners

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### Sponsors

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Ministry of High Education and Scientific Research, Iraq



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### Organizers

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University of Al-Qadisiyah, Iraq



Sunway University, Malaysia



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**Partners**

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University of Mosul, Iraq



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Mustansiriyah University, Iraq



Rochester Institute of Technology,  
Dubai



Warith Foundation for Technical  
Education, Iraq



Iraqi Engineers Union, Iraq



University of Kufa, Iraq



## Conference Programme

For latest schedule, please refer to  
<https://iceat2025.github.io/schedule>.



Venue: LT7, Sunway University Building

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### Day 1 (23<sup>rd</sup> July 2025)

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|       |   |                                                                                                                                    |
|-------|---|------------------------------------------------------------------------------------------------------------------------------------|
| 8:00  | ○ | Registration                                                                                                                       |
| 9:00  | ○ | Opening ceremony                                                                                                                   |
| 10:00 | ○ | Keynote 1<br><b>Advanced Materials for Energy Storage and Heat Transfer Applications</b><br>Prof. Dr. Saidur Rahman                |
| 10:45 | ○ | Tea break                                                                                                                          |
| 11:00 | ○ | Keynote 2<br><b>Toward 2035: Renewable Energy Innovations Transforming Our Future</b><br>AP. Dr. Hj. Firas Basim Alnaimi           |
| 11:45 | ○ | Keynote 3<br><b>Hydrogen Storage and Safety: Innovations, Challenges, and Industrial Applications</b><br>AP. Dr. Ali Sameer Muhsan |
| 12:30 | ○ | Lunch                                                                                                                              |
| 14:00 | ○ | Physical presentation                                                                                                              |
| 15:15 | ○ | Tea break                                                                                                                          |
| 15:30 | ○ | Physical presentation                                                                                                              |
| 17:00 | ○ | End of day 1                                                                                                                       |

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**Day 2 (24<sup>th</sup> July 2025)**

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|       |   |                                                                                                                                          |
|-------|---|------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30  | ○ | Keynote 4<br><b>Advances in Solar Photovoltaic Technology</b><br>Prof. Dr. Hussein A. Kazem                                              |
| 10:15 | ○ | Tea break                                                                                                                                |
| 10:45 | ○ | Keynote 5<br><b>Rubbercrete Bricks: Pioneering Sustainable Construction with Scrap Tire Innovation</b><br>Ir. AP. Dr. Bashar S. Mohammed |
| 11:30 | ○ | Keynote 6<br><b>Engineering Innovation: From Control to Creativity</b><br>Ir. Dr. Chua Yaw Long                                          |
| 12:15 | ○ | Lunch                                                                                                                                    |
| 14:00 | ○ | Campus tour                                                                                                                              |
| 15:15 | ○ | Tea break                                                                                                                                |
| 15:45 | ○ | Closing ceremony                                                                                                                         |
| 16:45 | ○ | End of day 2                                                                                                                             |

## Technical Sessions

For latest updates of technical sessions and participation links, please refer to <https://iceat2025.github.io/technicalsessions>.



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### Physical Presentation

Chaired by **Associate Professor Ir Dr Chua Bee Lin**

23 July, 2025 from 14:00 to 16:45

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|       |   |                                                                                                                                               |
|-------|---|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 14:15 | ○ | Paper ID 50<br>Numerical Investigation about Diesel Engine Powered by Waste Plastic Oil Blends under Different Load and Engine Speed          |
| 14:30 | ○ | Paper ID 117<br>A Metal-Free Electrocatalyst for Efficient Hydrogen Production                                                                |
| 14:45 | ○ | Paper ID 118<br>Development and Optimization of Recyclable GF-reinforced PLA Resin Composite for Renewable Energy Applications                |
| 15:00 | ○ | Paper ID 120<br>Bibliometric Analysis on Smart Self-Healing Nanocoating for 316L Stainless Steel Biomedical Implants                          |
| 15:15 | ○ | Break                                                                                                                                         |
| 15:30 | ○ | Paper ID 127<br>Effective Removal of Amoxicillin from Medical Wastewater Using an Eco-Friendly Modification of a Walnut Shell as an Adsorbent |
| 15:45 | ○ | Paper ID 132<br>Experimental and Computational Evaluation of Graphene-Enhanced Fiberglass/Polyester Nanocomposites                            |
| 16:00 | ○ | Paper ID 155<br>Buckling Analysis of Corrugated Plate Fuselage Under Uniform Pressure Loading Condition                                       |
| 16:15 | ○ | Paper ID 180<br>Application of Kaplan Meier estimator model for validation of AU-RKC as an early biomarker for Kidney Cancer                  |
| 16:30 | ○ | Paper ID 2<br>A Comprehensive Index Framework for Evaluating Urban Road Safety Performance: A Systematic Literature Review                    |

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Technical Session 1  
**Functional Materials and Nanotechnology in Chemical Engineering**

Chaired by **Professor Nishanth Gopalakrishnan Chemmangattuvalappil**  
23 July, 2025 from 10:00 to 12:00

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|       |                                                                                     |                                                                                                                                                  |
|-------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 |    | Paper ID 20<br>Improving The Properties of Thermoplastic Materials Using Polyester Resin With Nanoparticles: The State-Of-The-Art                |
| 10:15 |    | Paper ID 45<br>A Review of Novel Nanocomposite Adsorbents for Ultra-Deep Desulfurization of Diesel Fuel: Recent Advances and Future Perspectives |
| 10:30 |    | Paper ID 119<br>Impact of Nanomaterials Concentrations on the Stress–Strain Behavior and Elastic Modulus of Resin Coatings under an Applied Load |
| 10:45 |   | Paper ID 130<br>Optimization of CO <sub>2</sub> -Assisted Gravity Drainage Operational Parameters: Insights from a 2D Hele-Shaw Model            |
| 11:00 |  | Paper ID 168<br>Characterization and Properties of Reaction Majnoon Field rock stone with Carbon Dioxide                                         |
| 11:15 |  | Break                                                                                                                                            |
| 11:30 |  | Paper ID 175<br>Advanced Multiple Sclerosis Data Analysis Using Intensity-Guided Skull Removal and Level Set Method for Enhanced Accuracy        |
| 11:45 |  | Paper ID 205<br>A comprehensive overview of carbon dioxide separation technologies for post-combustion capture                                   |

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Technical Session 2  
**Intelligent Power and Communication Systems for Smart Infrastructure**

Chaired by **Ts. Dr Nor Akmar Mohd Yahya**  
23 July, 2025 from 10:00 to 12:15

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|       |                                                                                     |                                                                                                                          |
|-------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 10:00 |  | Paper ID 8<br>Three-Phase Seventeen-Level Cascaded Switched-Capacitor Multi-level Inverter for Grid-Connected PV Systems |
|-------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|



|       |   |                                                                                                                                              |
|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------|
| 10:15 | ○ | Paper ID 177<br>Safety-Oriented Design and Analysis of Dielectric Materials and Plate Geometry in Capacitive Wireless Power Transfer Systems |
| 10:30 | ○ | Paper ID 219<br>Evaluation and Enhancement of Solar Energy-Enabled Intelligent Structural Systems                                            |
| 10:45 | ○ | Paper ID 34<br>A High-Performance, Low-Cost Solution for Enhancing Capacity in Urban 5G Small Cells                                          |
| 11:00 | ○ | Paper ID 166<br>Design and Implementation of a Reconfigurable Dual Band Low Noise Amplifier for Modern Receiving Systems                     |
| 11:15 | ○ | Break                                                                                                                                        |
| 11:30 | ○ | Paper ID 112<br>Band Pass Filter Utilizing a Stepped Impedance Resonator for Modern Wireless Communication System                            |
| 11:45 | ○ | Paper ID 29<br>Investigation of Near Field Distributions of Meander Line Antenna in Human Body Conditions                                    |
| 12:00 | ○ | Paper ID 200<br>Bit Error Rate Analysis of Adaptive Modulation Techniques in Power Line Communication Channels                               |

### Technical Session 3 Composite Materials, Structural Me- chanics, and Manufacturing Innovations

Chaired by **Associate Professor Dr Anwar P.P. Abdul Majeed**  
23 July, 2025 from 10:00 to 12:15

|       |   |                                                                                                                   |
|-------|---|-------------------------------------------------------------------------------------------------------------------|
| 10:00 | ○ | Paper ID 170<br>A Review of Origami Structure According to Pattern, Manufactur-<br>ing and Application            |
| 10:15 | ○ | Paper ID 161<br>Study The Effect of Twin Wire Arc Spraying Parameters on Surface<br>Properties of Stainless Steel |
| 10:30 | ○ | Paper ID 154<br>An Analytical Review of Incremental Single Sheet Metal Forming<br>Techniques                      |
| 10:45 | ○ | Paper ID 150<br>Numerical analysis of multi –layer composite under free vibration<br>conditions                   |

|       |   |                                                                                                                                            |
|-------|---|--------------------------------------------------------------------------------------------------------------------------------------------|
| 11:00 | ○ | Paper ID 144<br>Effect of blades shape and duct geometrical parameters on Aero-dynamic performance of a small-sized axial fan              |
| 11:15 | ○ | Break                                                                                                                                      |
| 11:30 | ○ | Paper ID 109<br>Effect of Nano Silica Fillers on Dynamic Mechanical Performance and Accelerated Ageing Behavior of Carbon/Epoxy Composites |
| 11:45 | ○ | Paper ID 191<br>Designing a Structured Simulation and Project Based Learning Spine for Mechanical Engineering Education                    |
| 12:00 | ○ | Paper ID 213<br>Experimental Determination of the Influence of Perforation of Adherend to the adhesive bond strength                       |

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**Technical Session 4**  
**AI-Driven Intelligent Systems for Cyberse-**  
**curity, Healthcare, and Smart Infrastructure**

Chaired by **Dr Farrukh Hassan**  
 23 July, 2025 from 14:00 to 16:45

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|       |   |                                                                                                                           |
|-------|---|---------------------------------------------------------------------------------------------------------------------------|
| 14:00 | ○ | Paper ID 10<br>Optimizing Operating Room Scheduling Using Artificial Bee Colony and Bat Algorithm: A Comparative Analysis |
| 14:15 | ○ | Paper ID 25<br>SmartSARIMAX: An Advanced Model for Bandwidth Prediction in Data Networks                                  |
| 14:30 | ○ | Paper ID 221<br>Investigation of IMU Motion Effects based on MPU 6050 Using a Dynamic Motion Testing Device               |
| 14:45 | ○ | Paper ID 92<br>Review paper: Deep Learning Based Biometric Recognition Model Using Finger and Palm Vein Images            |
| 15:00 | ○ | Paper ID 104<br>Machine Learning-Driven Meta surfaces for Adaptive 6G Beam-forming in Dynamic Terahertz Channels          |
| 15:15 | ○ | Break                                                                                                                     |
| 15:30 | ○ | Paper ID 111<br>Predicting Adult Income Utilizing Various Artificial Intelligence Models                                  |

|       |   |                                                                                                                    |
|-------|---|--------------------------------------------------------------------------------------------------------------------|
| 15:45 | ○ | Paper ID 222<br>Designing a Robotic System for Efficient Exam Paper Distribution Using Multi-Sensor Technology     |
| 16:00 | ○ | Paper ID 193<br>Leveraging AI for Customer Segmentation and Predictive Insights to Elevate E-Commerce Satisfaction |
| 16:15 | ○ | Paper ID 201<br>PhisNet: Intelligent Detection of Phishing                                                         |
| 16:30 | ○ | Paper ID 202<br>A Real-Time Fall Detection Framework using Vision Transformer and LSTM for Elderly People          |

### Technical Session 5 **Mechanical Systems, Materials Engineering, and Thermal-Energy Applications**

Chaired by **Dr Shahrooz Eftekhari**  
23 July, 2025 from 14:00 to 04:45

|       |   |                                                                                                                                               |
|-------|---|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 14:00 | ○ | Paper ID 82<br>PMMA/PEEKs/MFP Ternary Composites Mechanical /Morphological Assessment for dental application.                                 |
| 14:15 | ○ | Paper ID 18<br>Surface Modification of Mg Alloys for Biomedical Applications Through the Electrospinning Process: A Review                    |
| 14:30 | ○ | Paper ID 105<br>Insights into Experimental and Numerical Evaluation of Liquid Desiccant Dehumidifier System Powered by Solar System           |
| 14:45 | ○ | Paper ID 80<br>Investigation of the Effect of Layer Number and Plate Perforation on the Ballistic Performance of Armor: A Review              |
| 15:00 | ○ | Paper ID 70<br>Numerical Investigation of Thermal - Hydraulic Performance of a Solar Air Heater Duct with transverse W-Shaped Rib Turbulators |
| 15:15 | ○ | Break                                                                                                                                         |
| 15:30 | ○ | Paper ID 31<br>Design and analysis of a planetary geared five-bar slider mechanism for generating long dwell periods                          |
| 15:45 | ○ | Paper ID 36<br>Study of 3D-Printed Honeycomb Orientation on Vibration Energy Harvesting                                                       |

|       |   |                                                                                                                                                          |
|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16:00 | ○ | Paper ID 178<br>Numerical analysis of perforation steel plates impacted by blunt, conical, hemispherical, and spherical projectiles                      |
| 16:15 | ○ | Paper ID 203<br>Evaluation of Mechanical and Thermal Properties of a Reinforced Thermosetting Polymer with Glass Fiber and Multi-Walled Carbon Nanotubes |
| 16:30 | ○ | Paper ID 216<br>Prosthetic Finger Design for Partial Hand Amputees: A Five-Bar Linkage Approach                                                          |

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Technical Session 6  
**Smart and Sustainable Materials  
in Civil Infrastructure Engineering**

Chaired by **Ts. Dr Tan Tee How**  
23 July, 2025 from 14:00 to 16:15

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|       |   |                                                                                                                                                                                     |
|-------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14:00 | ○ | Paper ID 137<br>The aesthetics of techniques in contemporary interior design (Kingdom of Jordan News Channel studio as a case study)                                                |
| 14:15 | ○ | Paper ID 147<br>An analysis of the environmental impacts of electronic waste and important precautions                                                                              |
| 14:30 | ○ | Paper ID 190<br>Reactive Powder Concrete Reinforced with Manufactured Fibers or Recycled from Waste Tire                                                                            |
| 14:45 | ○ | Paper ID 212<br>Behaviour of Fibre-Reinforced Lightweight Self-Compacting Concrete Containing Recycled Brick Aggregate and Silica Fume                                              |
| 15:00 | ○ | Paper ID 192<br>Valorization of Dredged Sediments from Cameron Highlands Reservoirs: Characterization and Pollution Risk Assessment for Sustainable Construction Materials Recovery |
| 15:15 | ○ | Break                                                                                                                                                                               |
| 15:30 | ○ | Paper ID 96<br>Experimental Investigation of Energy Performance in Double-Glazed Windows Utilizing Air and Phase Change Materials                                                   |
| 15:45 | ○ | Paper ID 165<br>A State-of-the-Art Review of Glazed Fenestration Systems for Enhanced Thermal Performance and Solar Heat Gain Mitigation in Building Envelopes                      |

- 16:00 ○ Paper ID 169  
Enhancement of wall insulation and reduction of heat gain in buildings by different ways: review

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**Technical Session 7**  
**Hydraulic and Urban Systems in Civil Engineering**

Chaired by **Associate Professor Ir Dr Ali Najah Ahmed Al Mahfoodh**  
23 July, 2025 from 14:00 to 16:30

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- 14:00 ○ Paper ID 57  
Structural Performance of High Strength Self-Compaction Concrete Beam with Perforated Steel Box Shear Connectors
- 14:15 ○ Paper ID 131  
Hydrodynamic Based Model to Simulate Water Quality in Kufa River, Iraq
- 14:30 ○ Paper ID 133  
Long Term Degradation of Tigris River at Al-Nuhairat Bridge Location
- 14:45 ○ Paper ID 135  
Experimental and numerical study of eco-friendly corbels under static loadin
- 15:00 ○ Paper ID 148  
Lead Removal from Water Using CTAB-Enhanced Nanosilica-Coated Sand Barrier under Continuous Flow: Experimental Study and Breakthrough Curve Modelling
- 15:15 ○ Break
- 15:30 ○ Paper ID 149  
Spatial analysis of noise pollution in the College of Engineering/ University of Basrah using (GIS) technology
- 15:45 ○ Paper ID 167  
Towards Sustainability in Urban Spaces: Evaluating Applied Architectural Ideas
- 16:00 ○ Paper ID 176  
Development of a Curve Number Map for the Ali Al-Gharbi District, Southern Iraq, Using GIS
- 16:15 ○ Paper ID 181  
Future Forecasts Of Rainfall Utilising The LARS-WG And CMIP6 Models

## Technical Session 8 Sustainable Structural Systems and Materials in Civil Engineering

Chaired by **Professor Ir Mo Kim Hung**  
24 July, 2025 from 10:00 to 12:45

|       |   |                                                                                                                                                                     |
|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 | ○ | Paper ID 55<br>Structural Behavior of One-Way Reinforced SCC Slabs Made with a Variety of Recycled Aggregates under Repeated Load                                   |
| 10:15 | ○ | Paper ID 61<br>Mechanical Properties of Modified Porous Concrete Incorporating Different Types of Lightweight Aggregate                                             |
| 10:30 | ○ | Paper ID 46<br>Flexural Behavior of Hybrid Concrete Tee Beams Reinforced with GFRP Bars                                                                             |
| 10:45 | ○ | Paper ID 49<br>Investigation the Efficacy of Steel Section Reinforcement in Hybrid Deep Beam Applications                                                           |
| 11:00 | ○ | Paper ID 11<br>Enhancing Metakaolin Reactivity in Geopolymers: A Comprehensive Review of Key Influencing Parameters                                                 |
| 11:15 | ○ | Break                                                                                                                                                               |
| 11:30 | ○ | Paper ID 138<br>Behavior of Recycled Aggregate Concrete Slab–Column Connection Strengthened by NSM GFRP bars                                                        |
| 11:45 | ○ | Paper ID 183<br>Influence of steel plate on enhancing the punching strength of voided slabs                                                                         |
| 12:00 | ○ | Paper ID 195<br>Criteria for Evaluating the Performance of Government Projects                                                                                      |
| 12:15 | ○ | Paper ID 210<br>Development of Sustainable Fibre Reinforced High Strength Lightweight Concrete with Recycled Fine Aggregate and Silica Fume                         |
| 12:30 | ○ | Paper ID 1<br>Application of Electro-Osmosis Method on Baghdad Soft Clay Shear Strength Using Vane Shear Test and Pile Installation Efficiency (Experimental Study) |

## Technical Session 9 Biofuels, Catalysis, and Resource Recovery

Chaired by **Ir Dr Yoon Li Wan**  
24 July, 2025 from 10:00 to 12:45

|       |                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 | Paper ID 15<br>Pyrolysis of Local Oil Seed for Biodiesel Production: A Review                                                                                                                            |
| 10:15 | Paper ID 44<br>Catalytic Deoxygenation of Hydrolyzed Animal Fat-Derived Oil Using a Zeolite Catalyst                                                                                                     |
| 10:30 | Paper ID 66<br>Enhancing Sulphur Removal from Iraqi Qayyarah Crude Oil through Glass Waste-Based Adsorption at Varied Temperatures and Contact Durations                                                 |
| 10:45 | Paper ID 77<br>Optimization of Thermal and Temporal Parameters in Ethanol-Assisted Soxhlet Extraction of Bioactive Drugs: A Kinetic Comparison of Eugenol and Caffeine                                   |
| 11:00 | Paper ID 89<br>Experimental Investigation and optimization for desulfurization of diesel fuel using microwave irradiation and activated carbon                                                           |
| 11:15 | Break                                                                                                                                                                                                    |
| 11:30 | Paper ID 94<br>Catalytic Deoxygenation of Hydrolyzed Oil of Beef Tallow over Lanthanum-Embedded HZSM-5 Zeolite Catalyst to Produce Bio-fuels                                                             |
| 11:45 | Paper ID 157<br>Synthesis and Characterization of Niobium promoted Ni/Mesoporous MCM-41 for potential application in steam reforming of glycerol using a fixed bed reactor                               |
| 12:00 | Paper ID 164<br>Elucidating CO <sub>2</sub> adsorptivity of Ni-Co-Y/MCM-41 as a potential pathway for CO <sub>2</sub> reforming of ethanol                                                               |
| 12:15 | Paper ID 189<br>Screening of hydrophilic deep eutectic solvents for ultrasound probe-assisted extraction of rosmarinic acid from salvia officinalis with high-performance liquid chromatography analysis |
| 12:30 | Paper ID 198<br>Efficient Glycerol Removal from Biodiesel Using Deep Eutectic Solvents Combined with Activated Carbon                                                                                    |

## Technical Session 10 Environmental and Wastewater Treatment Technologies

Chaired by **Ir Dr Ayu Haslija Abu Bakar**

24 July, 2025 from 10:00 to 12:45

|       |   |                                                                                                                                                               |
|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 | ○ | Paper ID 13<br>Treatment of petroleum refinery wastewater by advanced oxidation process (AOPs) Minireview                                                     |
| 10:15 | ○ | Paper ID 19<br>Treatment of Petroleum refinery wastewater by electrooxidation using anode composed of composite materials                                     |
| 10:30 | ○ | Paper ID 23<br>Assessment of Photocatalytic Process using composite photocatalysts for COD reduction for Al-Dewaniyah Petroleum refinery wastewater: RSM      |
| 10:45 | ○ | Paper ID 38<br>Hydrodynamic Characteristics of Two-Phase Systems in Tapered Bubble Columns (TBCs)                                                             |
| 11:00 | ○ | Paper ID 73<br>Equilibrium Analysis and Adsorptive Removal of Lead (Pb) by Activated Carbon Derived from Ceratophyllum demersum: Effect of Activation Methods |
| 11:15 | ○ | Break                                                                                                                                                         |
| 11:30 | ○ | Paper ID 97<br>Adsorptive desulphurization of model diesel by activated carbon loaded with Ni oxide nanoparticles prepared from local Iraqi pomegranate       |
| 11:45 | ○ | Paper ID 115<br>Modeling of the adsorption process of the organic pollutant from synthetic wastewater using different adsorbents                              |
| 12:00 | ○ | Paper ID 129<br>A Review of Natural Clay Ceramic Membranes: Manufacturing Techniques and Implementations in Industrial and Municipal Wastewater Treatment     |
| 12:15 | ○ | Paper ID 143<br>Kinetic model of Lead (Pb) Adsorption by Raw and Carbonized Ceratophyllum demersum: Mechanisms and Rate Analysis"                             |
| 12:30 | ○ | Paper ID 4<br>Ozone-Based Advanced Oxidation Processes for Dye Removal: A Brief Review                                                                        |



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**Technical Session 11**  
**Mechatronics, Robotics, and Smart Automation Systems**

Chaired by **Dr Richard Wong Teck Ken**  
24 July, 2025 from 10:00 to 11:30

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|-------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 |    | Paper ID 93<br>Evaluation and Enhancement of Solar Energy-Enabled Intelligent Structural Systems                                        |
| 10:15 |    | Paper ID 87<br>Investigation of IMU Motion Effects based on MPU 6050 Using a Dynamic Motion Testing Device                              |
| 10:30 |    | Paper ID 64<br>Designing a Robotic System for Efficient Exam Paper Distribution Using Multi-Sensor Technology                           |
| 10:45 |    | Paper ID 52<br>AI-Based Control of Brushless DC Motors: A Practical Approach to Performance Enhancement                                 |
| 11:00 |  | Paper ID 28<br>A Novel Coupled-Inductor Boost Converter with ANN Control for High Step-Up PV Systems                                    |
| 11:15 |  | Paper ID 174<br>Artificial Neural Network-Based Control of Vienna Rectifier for Power Factor Correction and Capacitor Voltage Balancing |

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**Technical Session 12**  
**Smart Computing, IoT Systems, and Intelligent Infrastructure**

Chaired by **Dr Nor Hafizah Binti Mohamed Halip**  
24 July, 2025 from 10:00 to 12:45

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|       |                                                                                     |                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 10:00 |  | Paper ID 35<br>Evolutionary Algorithms for Iot Service Selection: A Review                                                                |
| 10:15 |  | Paper ID 48<br>Intelligent Control for Video Broadcasting in Flying Ad Hoc Networks: A Simulation Study                                   |
| 10:30 |  | Paper ID 90<br>Patient-Specific 3D-Printed Cutting and Repositioning Guide for Mandibular Tumor Resection: Surgical Design and Validation |
| 10:45 |  | Paper ID 113<br>Swarm Intelligence in Modern Engineering A Comprehensive Review of Applications, Performance, and Emerging Trends         |

|       |                                                                                   |                                                                                                                                                          |
|-------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11:00 |  | Paper ID 136<br>Learning About Syndrome Awareness and WMS Algorithm for Adaptive Neural Decoding for 6G LDPC Base Graph Enhancement                      |
| 11:15 |  | Break                                                                                                                                                    |
| 11:30 |  | Paper ID 153<br>Design and Implementation of a Low Cost IoT-Based System for Enhanced Visual Feedback, Alert Systems, and Server Environment Tracking    |
| 11:45 |  | Paper ID 194<br>Leveraging Augmented and Virtual Reality for Enhancing Eco-Literacy: A Study on Sustainable Green Education Using Immersive Technologies |
| 12:00 |  | Paper ID 197<br>Automated Greenhouse Management System with IOT Integration                                                                              |
| 12:15 |  | Paper ID 14<br>Design and Implementation of an IoT-Based System for Remote Monitoring of Vital Signs                                                     |
| 12:30 |  | Paper ID 199<br>Developing a program to track the performance of multiple projects using Earned Value and Scorecard technology                           |

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**Main Hall-Second Campus**

24 July, 2025 from 09:00 to 12:30

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Paper ID 224

Laboratory Study on the Combined Effect of Novolac and SBS Polymers on the Physical and Mechanical Properties of Hot Mix Asphalt

Paper ID 225

Improving the Efficiency of the Biological Treatment of Hospitals' Wastewater by Using MBBR with k<sub>1</sub> and k<sub>3</sub> Biofilm Carriers

Paper ID 226

Comparison of the Capacity of a Roundabout and a Signalized Intersection

Paper ID 227

Cracking Resistance and Fracture Behavior of Modified Asphalt Mixtures

Paper ID 228

Improving Household Ceramic Filter Performance for Removing the Turbidity and Chlorine Using COMSOL

Paper ID 229

Development of a Nano-Sand Reactive Barrier for Lead Extraction: Performance Assessment in Synthetic and Real Groundwater Conditions

Paper ID 230

Enhancing the Mechanical Properties of polypropylene fiber-Reinforced High-Performance Concrete Using Attapulgitite-Based Internal Curing

Paper ID 231

Structural Behavior of Self-Compacting Reinforced Concrete One-Way Slab with Recycled Glass or Plastic as a Fine Aggregate Under Repeated Loads

Paper ID 232

Numerical Simulation of The Effect of Variation in Mechanical Properties of Concrete and Subgrade Enhanced with Iraqi Natural Zeolite on Rigid Pavement Response

Paper ID 233

A Comprehensive Index Framework for Evaluating Urban Road Safety Performance: A Systematic Literature Review

Paper ID 234

The Effect of High Range Water Reducing Admixture on Cement Content with Constant Compressive Strength and Workability

Paper ID 235

Environmental and Health Impacts of Medical Waste Shredding-Sterilization Technologies: A Review and Analysis using VOS viewer

Paper ID 236

Optimization of the Designed Water Network Distribution System Using MATLAB, Case Study Al-Diwaniyah

Paper ID 237

An Experimental Study of Surface Runoff Coefficients of Rainwater for Different Types of urban Roads: Sustainable Management

Paper ID 238

Evaluation of Sustainable Transport in the Urban Road Network: Public Transport Index for Baghdad City

Paper ID 239

Review Study on Back Calculation of Falling Weight Deflectometer (FWD) Test

Paper ID 240

A Review of the Semi-Flexible Pavement Applications and Effective Parameters

Paper ID 241

Reliability Analysis Assessment for Diwaniyah Water Treatment Plants as a Case Study

Paper ID 242

Review of Factors Affecting the Durability of Foamed Concrete

Paper ID 243

Development of Sustainable Self-Compacting Lightweight Precast Concrete Tiles incorporating Fine Recycled Brick Aggregates

Paper ID 244

The Mechanical and Physical Properties of Reactive Powder Concrete Reinforced with Carbon Nanotubes Reinforced with Standard Steel Fiber and Recycled Fiber from Tires

Paper ID 245

Bond Performance of Concrete-To-HPC Interface for Different Surface Profiles Using Slant and Bi-Surface Shear Test

Paper ID 246

Sustainable Production of Road Pavers using Recycled Plastic Waste

Paper ID 247

Key Factors Affecting the Mechanical Properties of Foamed Concrete: A Review

Paper ID 248  
Mechanical and Volumetric Changes of Sustainable Lightweight Fibrous Concrete with Fine Recycled Aggregate and Silica Fume

Paper ID 249  
Impact of Duct Position on Behavior of High-Strength Reinforced Concrete Flat Slab

Paper ID 250  
Laboratory Evaluation of Sustainable Asphalt Mix Using Bottom Ash and Styrene Butadiene Styrene (SBS)

Paper ID 251  
Electro-Fenton Oxidation for Degradation of Direct Red Organic Pollutants Using Stainless Steel and Iron Electrodes

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### Ordered by Paper ID

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|             |   |                       |   |               |
|-------------|---|-----------------------|---|---------------|
| Paper ID 1  | ○ | Technical Session 8   | ○ | 12:30 - 12:45 |
| Paper ID 2  | ○ | Physical Presentation | ○ | 16:30 - 16:45 |
| Paper ID 4  | ○ | Technical Session 10  | ○ | 12:30 - 12:45 |
| Paper ID 8  | ○ | Technical Session 2   | ○ | 10:00 - 10:15 |
| Paper ID 10 | ○ | Technical Session 4   | ○ | 14:00 - 14:15 |
| Paper ID 11 | ○ | Technical Session 8   | ○ | 11:00 - 11:30 |
| Paper ID 13 | ○ | Technical Session 10  | ○ | 10:00 - 10:15 |
| Paper ID 14 | ○ | Technical Session 12  | ○ | 12:15 - 12:30 |
| Paper ID 15 | ○ | Technical Session 9   | ○ | 10:00 - 10:15 |
| Paper ID 18 | ○ | Technical Session 5   | ○ | 14:15 - 14:30 |
| Paper ID 19 | ○ | Technical Session 10  | ○ | 10:15 - 10:30 |
| Paper ID 20 | ○ | Technical Session 1   | ○ | 10:00 - 10:15 |
| Paper ID 23 | ○ | Technical Session 10  | ○ | 10:30 - 10:45 |
| Paper ID 25 | ○ | Technical Session 4   | ○ | 14:15 - 14:30 |
| Paper ID 28 | ○ | Technical Session 11  | ○ | 11:00 - 11:15 |
| Paper ID 29 | ○ | Technical Session 2   | ○ | 11:45 - 12:00 |
| Paper ID 31 | ○ | Technical Session 5   | ○ | 15:30 - 15:45 |
| Paper ID 34 | ○ | Technical Session 2   | ○ | 10:45 - 11:00 |
| Paper ID 35 | ○ | Technical Session 12  | ○ | 10:00 - 10:15 |
| Paper ID 36 | ○ | Technical Session 5   | ○ | 15:45 - 16:00 |
| Paper ID 38 | ○ | Technical Session 10  | ○ | 10:45 - 11:00 |
| Paper ID 44 | ○ | Technical Session 9   | ○ | 10:15 - 10:30 |
| Paper ID 45 | ○ | Technical Session 1   | ○ | 10:15 - 10:30 |
| Paper ID 46 | ○ | Technical Session 8   | ○ | 10:30 - 10:45 |
| Paper ID 48 | ○ | Technical Session 12  | ○ | 10:15 - 10:30 |
| Paper ID 49 | ○ | Technical Session 8   | ○ | 10:45 - 11:00 |
| Paper ID 50 | ○ | Physical Presentation | ○ | 14:15 - 14:30 |
| Paper ID 52 | ○ | Technical Session 11  | ○ | 10:45 - 11:00 |
| Paper ID 55 | ○ | Technical Session 8   | ○ | 10:00 - 10:15 |

## Technical Sessions

|              |   |                       |   |               |
|--------------|---|-----------------------|---|---------------|
| Paper ID 57  | ○ | Technical Session 7   | ○ | 14:00 - 14:15 |
| Paper ID 61  | ○ | Technical Session 8   | ○ | 10:15 - 10:30 |
| Paper ID 64  | ○ | Technical Session 11  | ○ | 10:30 - 10:45 |
| Paper ID 66  | ○ | Technical Session 9   | ○ | 10:30 - 10:45 |
| Paper ID 70  | ○ | Technical Session 5   | ○ | 15:00 - 15:30 |
| Paper ID 73  | ○ | Technical Session 10  | ○ | 11:00 - 11:15 |
| Paper ID 74  | ○ | Technical Session 4   | ○ | 14:30 - 14:45 |
| Paper ID 77  | ○ | Technical Session 9   | ○ | 10:45 - 11:00 |
| Paper ID 80  | ○ | Technical Session 5   | ○ | 14:45 - 15:00 |
| Paper ID 82  | ○ | Technical Session 5   | ○ | 14:00 - 14:15 |
| Paper ID 87  | ○ | Technical Session 11  | ○ | 10:15 - 10:30 |
| Paper ID 89  | ○ | Technical Session 9   | ○ | 11:00 - 11:15 |
| Paper ID 90  | ○ | Technical Session 12  | ○ | 10:30 - 10:45 |
| Paper ID 92  | ○ | Technical Session 4   | ○ | 14:45 - 15:00 |
| Paper ID 93  | ○ | Technical Session 11  | ○ | 10:00 - 10:15 |
| Paper ID 94  | ○ | Technical Session 9   | ○ | 11:30 - 11:45 |
| Paper ID 96  | ○ | Technical Session 6   | ○ | 15:30 - 15:45 |
| Paper ID 97  | ○ | Technical Session 10  | ○ | 11:30 - 11:45 |
| Paper ID 104 | ○ | Technical Session 4   | ○ | 15:00 - 15:15 |
| Paper ID 105 | ○ | Technical Session 5   | ○ | 14:30 - 14:45 |
| Paper ID 109 | ○ | Technical Session 3   | ○ | 11:30 - 11:45 |
| Paper ID 111 | ○ | Technical Session 4   | ○ | 15:30 - 15:45 |
| Paper ID 112 | ○ | Technical Session 2   | ○ | 11:30 - 11:45 |
| Paper ID 113 | ○ | Technical Session 12  | ○ | 10:45 - 11:00 |
| Paper ID 115 | ○ | Technical Session 10  | ○ | 11:45 - 12:00 |
| Paper ID 117 | ○ | Physical Presentation | ○ | 14:30 - 14:45 |
| Paper ID 118 | ○ | Physical Presentation | ○ | 14:45 - 15:00 |
| Paper ID 119 | ○ | Technical Session 1   | ○ | 10:30 - 10:45 |
| Paper ID 120 | ○ | Physical Presentation | ○ | 15:00 - 15:15 |
| Paper ID 127 | ○ | Physical Presentation | ○ | 15:30 - 15:45 |
| Paper ID 129 | ○ | Technical Session 10  | ○ | 12:00 - 12:15 |
| Paper ID 130 | ○ | Technical Session 1   | ○ | 10:45 - 11:00 |
| Paper ID 131 | ○ | Technical Session 7   | ○ | 14:15 - 14:30 |
| Paper ID 132 | ○ | Physical Presentation | ○ | 15:45 - 16:00 |
| Paper ID 133 | ○ | Technical Session 7   | ○ | 14:30 - 14:45 |
| Paper ID 135 | ○ | Technical Session 7   | ○ | 14:45 - 15:00 |
| Paper ID 136 | ○ | Technical Session 12  | ○ | 11:00 - 11:15 |
| Paper ID 137 | ○ | Technical Session 6   | ○ | 14:00 - 14:15 |
| Paper ID 138 | ○ | Technical Session 8   | ○ | 11:30 - 11:45 |
| Paper ID 143 | ○ | Technical Session 10  | ○ | 12:15 - 12:30 |
| Paper ID 144 | ○ | Technical Session 3   | ○ | 11:00 - 11:15 |
| Paper ID 148 | ○ | Technical Session 7   | ○ | 15:00 - 15:15 |
| Paper ID 149 | ○ | Technical Session 7   | ○ | 15:30 - 15:45 |
| Paper ID 150 | ○ | Technical Session 3   | ○ | 10:45 - 11:00 |

## Technical Sessions

|              |   |                       |   |               |
|--------------|---|-----------------------|---|---------------|
| Paper ID 153 | ○ | Technical Session 12  | ○ | 11:30 - 11:45 |
| Paper ID 154 | ○ | Technical Session 3   | ○ | 10:30 - 10:45 |
| Paper ID 155 | ○ | Physical Presentation | ○ | 16:00 - 16:15 |
| Paper ID 157 | ○ | Technical Session 9   | ○ | 11:45 - 12:00 |
| Paper ID 161 | ○ | Technical Session 3   | ○ | 10:15 - 10:30 |
| Paper ID 164 | ○ | Technical Session 9   | ○ | 12:00 - 12:15 |
| Paper ID 165 | ○ | Technical Session 6   | ○ | 15:45 - 16:00 |
| Paper ID 166 | ○ | Technical Session 2   | ○ | 11:00 - 11:15 |
| Paper ID 167 | ○ | Technical Session 7   | ○ | 15:45 - 16:00 |
| Paper ID 168 | ○ | Technical Session 1   | ○ | 11:00 - 11:15 |
| Paper ID 169 | ○ | Technical Session 6   | ○ | 16:00 - 16:15 |
| Paper ID 170 | ○ | Technical Session 3   | ○ | 10:00 - 10:15 |
| Paper ID 174 | ○ | Technical Session 11  | ○ | 11:15 - 11:30 |
| Paper ID 175 | ○ | Technical Session 1   | ○ | 11:30 - 11:45 |
| Paper ID 176 | ○ | Technical Session 7   | ○ | 16:00 - 16:15 |
| Paper ID 177 | ○ | Technical Session 2   | ○ | 10:15 - 10:30 |
| Paper ID 178 | ○ | Technical Session 5   | ○ | 16:00 - 16:15 |
| Paper ID 180 | ○ | Physical Presentation | ○ | 16:15 - 16:30 |
| Paper ID 181 | ○ | Technical Session 7   | ○ | 16:15 - 16:30 |
| Paper ID 183 | ○ | Technical Session 8   | ○ | 11:45 - 12:00 |
| Paper ID 189 | ○ | Technical Session 9   | ○ | 12:15 - 12:30 |
| Paper ID 190 | ○ | Technical Session 6   | ○ | 14:30 - 14:45 |
| Paper ID 191 | ○ | Technical Session 3   | ○ | 11:45 - 12:00 |
| Paper ID 192 | ○ | Technical Session 6   | ○ | 15:00 - 15:15 |
| Paper ID 193 | ○ | Technical Session 4   | ○ | 16:00 - 16:15 |
| Paper ID 194 | ○ | Technical Session 12  | ○ | 11:45 - 12:00 |
| Paper ID 195 | ○ | Technical Session 8   | ○ | 12:00 - 12:15 |
| Paper ID 197 | ○ | Technical Session 12  | ○ | 12:00 - 12:15 |
| Paper ID 198 | ○ | Technical Session 9   | ○ | 12:30 - 12:45 |
| Paper ID 199 | ○ | Technical Session 12  | ○ | 12:30 - 12:45 |
| Paper ID 200 | ○ | Technical Session 2   | ○ | 12:00 - 12:15 |
| Paper ID 201 | ○ | Technical Session 4   | ○ | 16:15 - 16:30 |
| Paper ID 202 | ○ | Technical Session 4   | ○ | 16:30 - 16:45 |
| Paper ID 203 | ○ | Technical Session 5   | ○ | 16:15 - 16:30 |
| Paper ID 205 | ○ | Technical Session 1   | ○ | 11:45 - 12:00 |
| Paper ID 210 | ○ | Technical Session 8   | ○ | 12:15 - 12:30 |
| Paper ID 212 | ○ | Technical Session 6   | ○ | 14:45 - 15:00 |
| Paper ID 213 | ○ | Technical Session 3   | ○ | 12:00 - 12:15 |
| Paper ID 216 | ○ | Technical Session 5   | ○ | 16:30 - 16:45 |
| Paper ID 219 | ○ | Technical Session 2   | ○ | 10:30 - 10:45 |
| Paper ID 221 | ○ | Technical Session 4   | ○ | 14:30 - 14:45 |
| Paper ID 222 | ○ | Technical Session 4   | ○ | 15:45 - 16:00 |
| Paper ID 224 | ○ | Poster – Main Hall    | ○ | 10:00 - 12:30 |
| Paper ID 225 | ○ | Poster – Main Hall    | ○ | 10:00 - 12:30 |
| Paper ID 226 | ○ | Poster – Main Hall    | ○ | 10:00 - 12:30 |

## Technical Sessions

|              |   |                    |   |               |
|--------------|---|--------------------|---|---------------|
| Paper ID 227 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 228 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 229 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 230 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 231 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 232 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 233 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 234 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 235 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 236 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 237 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 238 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 239 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 240 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 241 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 242 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 243 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 244 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 245 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 246 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 247 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 248 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 249 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 250 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |
| Paper ID 251 | ○ | Poster – Main Hall | ○ | 10:00 - 12:30 |



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## Keynote Speakers

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### Prof. Dr. Saidur Rahman

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Professor Saidur Rahman is currently working as a Distinguished Research Professor and Head of the Research Centre for Nano-Materials and Energy Technology (RCNMET) at Sunway University. He is also working with Lancaster University as a full Professor. Previously, he worked as a Chair Professor at the Center of Research Excellence in Renewable Energy at King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. Prior to joining KFUPM, Prof. Saidur worked 18 years in University of Malaya, a premier University in Malaysia. He is ranked 1 in Malaysia according to top 2% Scientist by Stanford/Elsevier Analysis. He is also number 1 Scientist in Malaysia according to AD scientific, research.com and GPS scholar. Clarivate Analytics/Thomson Reuters awarded him highly cited researcher for being among the top 1% researchers for most cited documents in his research field for the eight consecutive years (2014-2021). In 2019 and 2024, Prof. Saidur won Vice Chancellor's award for achievement in Research, Sunway University. Prof. Saidur published more than 900 journals and conference papers. Majority of them are in top ranking high impact journals. His publications are cited more than 77,000 times with an h-index of 139 according to Google Scholar citation. He has supervised more than 80 postgraduate students so far and has secured and managed more than 25 million ringgit research grants as a PI and member. Prof. Saidur is working in the area of emerging nano-materials (MXenes) and their applications in energy storage, heat transfer, solar energy harvesting and environmental remediation.

#### Speech Sessions

Advanced Materials for energy storage and heat transfer applications

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**AP. Dr. Hj. Firas Basim Alnaimi**

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Dr. Firas Basim Ismail, Associate Professor and Head of the Smart Power Generation Research Center, Universiti Tenaga Nasional (UNITEN), Malaysia. Adjunct Professor, Sohar University, Oman. (Concurrent post)

Dr. Firas Basim Ismail holds a Ph.D. in Mechanical Engineering from Universiti Teknologi PETRONAS, Malaysia (2010), an M.Sc. in Mechanical Engineering from the University of Technology, Iraq (2003), and a B.Sc. in Mechanical Engineering from the University of Technology, Iraq (1999). He completed his secondary education at Baghdad College High School—American School, Baghdad, Iraq (1995).

He joined Universiti Tenaga Nasional in 2013, where he has been actively engaged in research and development in thermofluid dynamics, power plant performance optimization, and renewable-smart power generation technologies. He has successfully led over 25 research projects, authored more than 150 journal and conference papers, and received 61 international awards. Dr. Firas is a chartered engineer with IMechE, UK, and has been invited as a plenary speaker at several international conferences.

**Speech Sessions**

Toward 2035: Renewable Energy Innovations Transforming Our Future

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**AP. Dr. Ali Sameer Muhsan**

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Dr. Ali S. Muhsan is a Senior Lecturer at the Mechanical Engineering Department, Universiti Teknologi PETRONAS (UTP), Malaysia. Dr. Ali Sameer Muhsan earned his PhD in 2014 from Universiti Teknologi PETRONAS. He completed his master's degree in 2010 at Universiti Sains Malaysia. In 2006, he received his bachelor's degree from Al-Mustansiriya University in Baghdad, Iraq.

His expertise lies in nanotechnology, advanced materials, and manufacturing technology, with a strong focus on Oil & Gas applications, including proppant coating technologies, enhanced oil recovery (EOR), and drilling fluid additives. He has contributed significantly to hydrogen storage systems, thermal management, and nanomaterial development for energy applications. With over 75 high-impact publications, multiple patents, and industry collaborations, his work has led to innovative solutions for hydraulic fracturing, thermal conductivity enhancement, and energy-efficient materials. Dr. Muhsan has secured over RM2 million in research funding and has been actively involved in industry-academia partnerships, consulting on hydrogen safety, nanotechnology, and energy-efficient solutions.

**Speech Sessions**

Hydrogen Storage and Safety: Innovations, Challenges, and Industrial Applications

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**Prof. Dr. Hussein A. Kazem**

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Professor Dr Hussein A. Kazem holds a BSc and MSc in electrical engineering from the University of Technology (UOT), Baghdad-Iraq, and a PhD from Newcastle University (NCL), United Kingdom. His academic journey began in 1995 at Al-Mamon College, and in 1996 he assumed the role of Assistant Lecturer at UOT. From 1997 to 2002, he served as a Program Coordinator at the Faculty of Engineering, Al Tahady University, Libya. He joined Sohar University (SU) in 2002. Hussein is a Professor at SU, Oman. He also holds the position of a visiting Professor at UKM and UNITEN-Malaysia, and Newcastle University-UK.

With a dedicated focus on academics and research spanning over 25 years, Hussein is actively engaged with prestigious professional organizations and engineering societies. He serves as a member, researcher, editor, and reviewer several publishers. Hussein has organized and participated in numerous conferences, symposiums, and workshops. His scholarly contributions include more than 270 published papers in scientific journals and conferences, 60 invited talks, nine chapters, and eight books. Recognizing his exceptional research, Hussein has been honored with the SU Vice- Chancellor Award for Research, as well as national and international accolades such as the “Golden Medal Award” at Pecipta’13 in Malaysia, “The Outstanding Renewable Energy Lab Award” at the World Renewable Energy Congress 2014 in UK, “Renewable & Sustainable Energy Pioneer Award” at the World Renewable Energy Congress 2016 in Indonesia, and “Special Award of Excellence in Renewable Energy Research” at the World Renewable Energy Congress 2018 in Kingston-UK. In addition, he has received the IEEE Best Paper Award in 2021, Golden Medal Award on Malaysia Technology Expo (MTE) 2023, Golden Medal Award and Special Award on MTE 2024, and holds two patents as an inventor and co-inventor. Hussein has successfully supervised and guided the graduation of more than 60 BSc, 11 MSc, and 9 PhD students.

Throughout his career, Hussein has led and contributed to several research projects, including six research grants. His research interests revolve around Photovoltaic, solar thermal, PV/T, Renewable Energy, Power Electronics, Power Quality, and Electrical Power Systems. Notably, Hussein serves as the chairman of the Renewable Energy & Sustainable Technology Research Group and leads the Generation & Storage task force in the Oman Renewable Energy Strategic Program. Furthermore, he holds the position of Associate Editor-in-Chief for the journal of Solar Energy.

**Speech Sessions**

Advances in Solar Photovoltaic Technology

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**Ir. AP. Dr. Bashar S. Mohammed**

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Ir. Dr. Bashar S. Mohammed currently serves as an Associate Professor in the Civil and Environmental Engineering Department at Universiti Teknologi PETRONAS. With a profound expertise in civil and construction engineering, his focus lies in the development of sustainable materials, particularly rubbercrete, geopolymer interlocking bricks, and engineered cementitious composites. Dr. Bashar has demonstrated exceptional leadership, notably as the former Chair of the Civil and Environmental Engineering Department and as Head of the Off-Shore Engineering Centre of Excellence at UTP. His impressive credentials, which include being a Registered Professional Engineer (BEM), Fellow and Chartered Engineer (ICE), and Project Management Professional (PMP®), reflect his unparalleled technical proficiency. Renowned for his ability to promote collaboration between global leaders, Dr. Bashar has chaired multiple international engineering conferences and served as President of the Mygeopolymer Society. His academic and leadership achievements are complemented by over 194 SCOPUS-indexed publications and numerous international honours, including multiple gold awards in prestigious innovation exhibitions. Moreover, he has authored six influential books, among them the award-winning “Rubbercrete Interlocking Bricks,” which received the Best Scientific Book Award 2021 from the Malaysian Scientific Publishing Council, as well as “Design of Reinforced Concrete Elements to Eurocode 2,” which honoured with the National Book Award 2018 by the National Book Development Foundation, Malaysia.

**Speech Sessions**

Rubbercrete Bricks: Pioneering Sustainable Construction with Scrap Tire Innovation

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**Ir. Dr. Chua Yaw Long**

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Ir. Dr. Chua Yaw Long has 22 years of teaching experience, including transnational education involving HEIs in Malaysia, UK, and Dubai. His major expertise are Control System Engineering and Engineering Design Process. He has taught a wide range of modules for undergraduates and postgraduates, in-person and virtually. He has successfully mentored colleagues to be registered as Chartered Engineers with the Engineering Council as well as Professional Engineer with the Board of Engineers Malaysia. His research focus includes capacity building (Creativity and Thinking Skills, Learning Innovation, Complex Problem- Solving Skills) and engineering related areas (Robotics, Mechatronics, Automation, Control and Renewable Energy) to bring a new paradigm shift in engineering education-social sciences studies. He has supervised over 150 research projects in various areas in STEM and Engineering Education, Capacity Building, Creativity and Innovation Engineering. He has published 55 peer-reviewed journals and proceedings.

**Speech Sessions**

Engineering Innovation: From Control to Creativity

## Committees

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### Chairmen

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1. Prof. Dr. Ali A. Jazie, University of Al-Qadisiyah, Iraq.
2. Prof. Ir. Dr. Denny Ng Kok Sum, Sunway University, Malaysia.

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### Steering Committee

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1. Prof. Dr. Haitham Ali Bady, University of Al-Qadisiyah, Iraq.
2. AP Dr. Hoo Choon Lih, Sunway University, Malaysia.
3. Prof. Dr. Malik Jasim Farhan, Mustansiriyah University, Iraq.
4. Prof. Dr. Yousef Al Assaf, Rochester Institute of Technology of Dubai.
5. Prof. Dr. A.I. Al-Hadidy, University of Mosul, Iraq.

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### Organizing Committee

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1. Salih A. Rushdi, PhD, University of Al-Qadisiyah.
2. Richard Wong Teck Ken, PhD, Sunway University.
3. Husam Kareem Mohsin Al-Jothery, PhD, University of Al-Qadisiyah.
4. Wisam Jasim Kadhim Al-obaidi, PhD, University of Al-Qadisiyah.
5. Chua Bee Lin, PhD, Sunway University.
6. Hayder Miri Hamzah, PhD, University of Al-Qadisiyah.
7. Ali Najah Ahmed Al Mahfoodh, PhD, Sunway University.
8. Hasan Hamid Naji, PhD, University of Al-Qadisiyah.
9. Asaad Sasaa Agrab, MSc, University of Al-Qadisiyah.
10. Foo Jinny, MSc, Sunway University.

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### Scientific Committee

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1. Mohamed F. Al-Dawody, PhD, University of AL-Qadisiyah.(Chair)
2. Ali H. Abbar, PhD, University of Baghdad.
3. Isidro de Jesús Sánchez-Arce, PhD, Universidade de Trás-os-Montes e Alto Douro, Portugal. (Guest Editor)
4. Muhanad Dheyaa Hashim, PhD, University of AL-Qadisiyah.
5. Thair Shakir Mahmoud, PhD, Edith Cowan University, Australia. (Guest Editor)

6. Muhieddin Amer, PhD, Rochester Institute of Technology of Dubai, UAE.
7. Rafid Kadhim Abbas, PhD, University of AL-Qadisiyah.
8. Majid H. Majeed, PhD, University of Al-Mustaqbal.
9. Richard Wong Teck Ken, PhD, Sunway University, Malaysia.
10. Ali Samer Muhsan, PhD, Universiti Teknologi PETRONAS, Malaysia. (Guest Editor)
11. Dhafer A. Hamzah, PhD, University of AL-Qadisiyah.
12. Wael A. Samad, PhD, Rochester Institute of Technology of Dubai, UAE.
13. Mohammad Shakir Nasif, PhD, Universiti Teknologi PETRONAS, Malaysia.
14. Mohammed Saleh Mohammed, PhD, University of Mosul.
15. Ali Najah Ahmed Al Mahfoodh, PhD, Sunway University, Malaysia.
16. Sofyan Younis Kashmola, PhD, University of Mosul. (Guest Editor)
17. Hoo Choon Lih, PhD, Sunway University, Malaysia.
18. Dhuha Abdulgani Al-Kazzaz, PhD, University of Mosul.
19. Aawag Mohsen Alawag, PhD, Universiti Teknologi MARA, Malaysia. (Guest Editor)
20. Thar M. Albarody, PhD, University of Technology.
21. Najib Al-Shaibani, PhD, University of Vermont, USA.
22. Mazen Makki Ali, PhD, University of AL-Qadisiyah.

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### **IT Coordinators**

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1. Osama Majeed Hilal Almiahi, PhD.
2. Mubeen Isam, MSc.
3. Ahmed Thabit Abed Al-Rubaye, MSc.
4. Hasanain Mohammed Jaafar, Eng.



